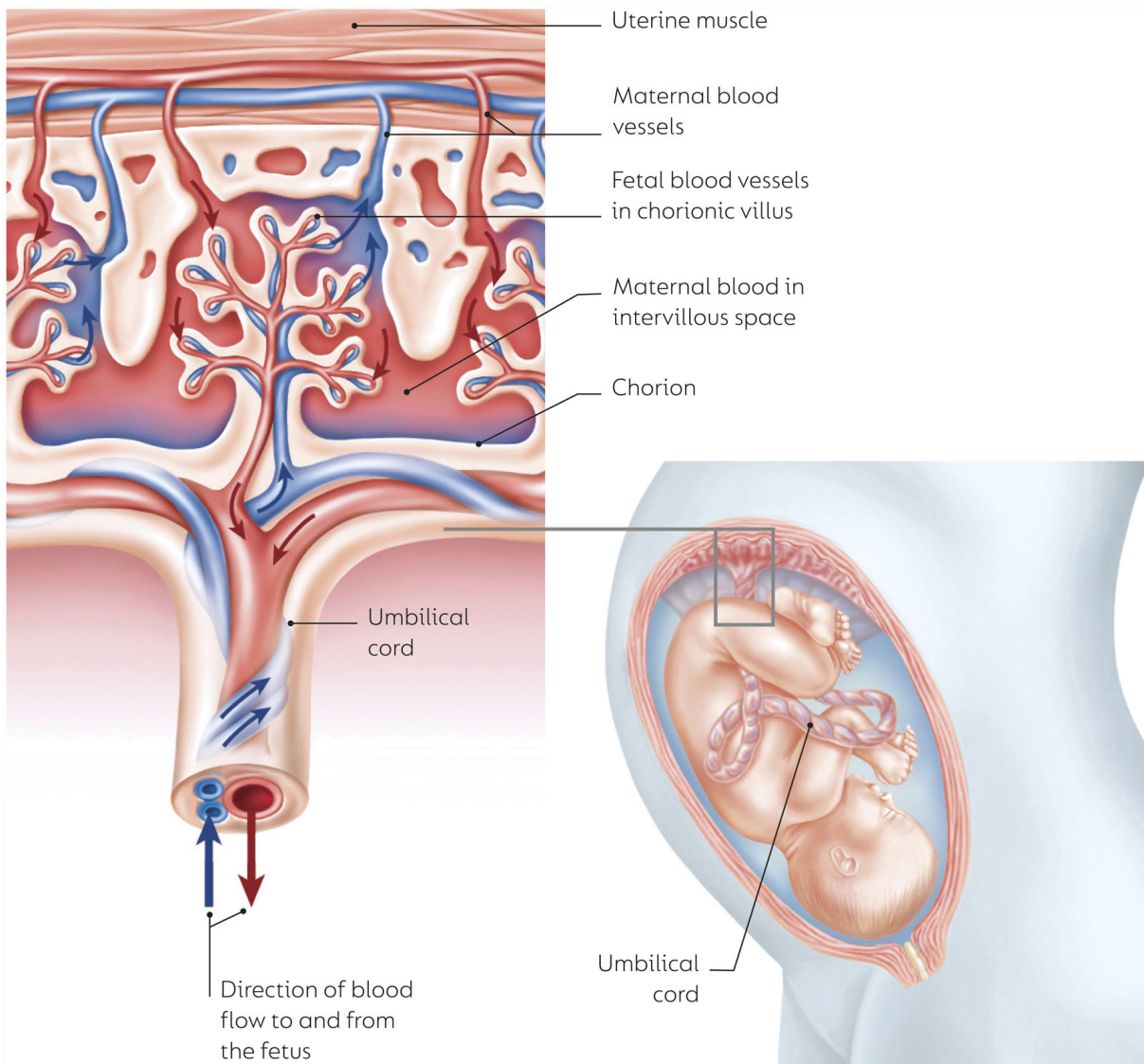




The vast network of villi in the placenta facilitates the transfer of gases, nutrients, and waste products to and from your baby.

The role of the placenta

The placenta fulfills many essential roles that help sustain the pregnancy and enable the baby to develop.

The exchange of substances. The placenta transports substances to and from your baby, acting as your baby's lungs, kidneys, and digestive system.

To obtain oxygen, the baby's blood cells grab oxygen molecules from your own hemoglobin (the oxygen-carrying substance in blood). Fetal hemoglobin has a modified structure that makes it bind readily to oxygen. Your baby requires twice as much oxygen weight for weight, as you, so the transfer of oxygen needs to be efficient. The placenta's enhanced blood supply, large surface area, and the characteristics of fetal hemoglobin all ensure the efficient transfer of oxygen from mother to baby. As your hemoglobin gives up oxygen, it accepts carbon dioxide molecules. Your lungs breathe